

**Multiple Choice:**

1. The purpose of the fetch cycle in a computer is to:
  - A. Obtain input data
  - B. Obtain memory data
  - C. Obtain an instruction
  - D. Implement a specific operation
2. A three-state buffer is a device that:
  - A. Has three different output lines
  - B. Has three different logic levels
  - C. Has a high impedance state in addition to normal 0 and 1 levels
  - D. Is a special type of storage register
3. A sequential logic system has outputs that depend on:
  - A. Only the inputs to the system at the current time
  - B. Present as well as past inputs
  - C. Memory device outputs independent of system inputs
  - D. None of these
4. A static memory will store information:
  - A. Even when power is not applied to the memory
  - B. As long as power is applied to the memory
  - C. As long as power is applied and the memory is refreshed periodically
  - D. None of these
5. Clock periods are measured from:
  - A. Similar points on the clock waveform
  - B. From the low level to the high level
  - C. From the high level to the low level
  - D. From the time the clock pulse is at 50% of its low-to-high transition until it is at 50% of its high-to-low transition
6. The rise time of a signal is the time:
  - A. To make a transition from a low level to a high level
  - B. To make a transition from a high level to a low level
  - C. For the output signal to start making a transition to a high level from the time an input signal causes the transition to occur
  - D. Between midpoints of a square wave signal
7. The purpose of the logic analyzer is to:
  - A. Verify the logic operation of the gates in a system
  - B. Sample and display system signals
  - C. Analyze the functional logic operations of a system
  - D. None of these
8. If a NAND latch has a 1 on the SET input and 0 on the CLEAR input, then the SET input goes to 0, the latch will be:
  - A. HIGH
  - B. LOW
  - C. Invalid
  - D. None of these
9. The fan-in of a logic gate refers to the number of
  - A. input devices that can be connected
  - B. input terminals
  - C. output terminals
  - D. circuits output can drive
10. In digital circuits Schottky transistors are preferred over normal transistor because of their
  - A. lower propagation delay
  - B. higher propagation delay
  - C. lower power dissipation
  - D. higher power dissipation
11. A dual threshold display in a logic analyzer provides information about:
  - A. Glitch occurrences
  - B. Rise times
  - C. Fall times
  - D. B and C above
12. A logic gate is an electronic circuit which
  - A. makes logic decision
  - B. allows electron flow only in one direction
  - C. works on binary data
  - D. alternates between 0 and 1 values
13. The logic gate which produces a 0 or low-level output when one or both of the inputs are 1 is called \_\_\_\_\_ gate.
  - A. AND
  - B. NOR
  - C. OR
  - D. NAND
14. The hold time for a D-type flip-flop is the time the:
  - A. Data will be held on the output after a clock signal
  - B. Data must be held on the input after a clock edge
  - C. Clock must be high
  - D. Clock must be low
15. A NOR gate is ON only when all its inputs are
  - A. ON
  - B. High
  - C. Positive
  - D. OFF
16. The first person who used Boolean algebra for the design of relay switching circuits was
  - A. Aristotle
  - B. Boole
  - C. Shannon
  - D. Ramanujam
17. An XOR gate produces an output only when its two inputs as
  - A. high
  - B. different
  - C. low
  - D. same
18. While obtaining minimal sum of product expression,
  - A. all don't cares are ignore



- B. all don't cares are treated as logic 1s  
C. all don't cares are treated as logic 0s  
D. only such don't cares that aid minimization are treated as logic 1s
19. Boolean algebra is essential based on  
A. symbols C. logic  
B. truth D. numbers
20. The most obvious identifying feature of a TTL gate is its  
A. large fan-out  
B. high power dissipation  
C. interconnected transistors  
D. multiemitter transistor
21. A 555 IC timer can be used to operate as  
A. a monostable multivibrator  
B. an astable multivibrator  
C. a voltage controlled oscillator  
D. All of these
22. A decoder is an MSI circuit whose output lines are controlled by:  
A. A binary input select code  
B. A similar set of input lines  
C. An output enable  
D. A clock input
23. A dynamic memory generally contains:  
A. No decoders  
B. Row decoders  
C. Column decoders  
D. Both row and column decoders
24. The purpose of a clock input to a flip-flop is to  
A. clear the device  
B. set the device  
C. always cause the output to change the states  
D. cause the output to assume a state dependent on the controlling inputs
25. The digital equivalent of an electric series circuit is the \_\_\_\_ gate.  
A. NOR C. NAND  
B. OR D. AND
26. A half-adder can be constructed from  
A. two XNOR gates only  
B. one XOR and one OR gate with their outputs connected in parallel  
C. one XOR and one OR gate with their inputs connected in parallel  
D. one XOR gate and one AND gate
27. A unique operating feature of ECL circuit is its  
A. very high speed  
B. high power dissipation  
C. series base resistor  
D. compatibility with other logic series
28. For getting an output from an XNOR gate, its both inputs must be  
A. high  
B. low  
C. at the same logic level  
D. at the opposite logic levels
29. A glitch is a:  
A. Momentary 0-to-1-to-0 signal sequence  
B. Momentary 1-to-0-to-1 signal sequence  
C. A noise pulse  
D. Any of the above
30. Rise and fall times are measured from:  
A. 50% points on a waveform  
B. 0 voltage level time to 1 voltage level time  
C. 10% to 90% points on the voltage transition  
D. The input signal to the output signal
31. The processor in a computer performs:  
A. The function of main system memory  
B. Involves transfer of data only from memory to processor or back  
C. Provides only arithmetic or logic operations  
D. Is the central control element of the system
32. A dynamic memory will store information:  
A. Even when power is not applied to the memory  
B. As long as power is applied to the memory  
C. As long as power is applied and the memory is refreshed periodically  
D. None of these
33. To cause a three-state buffer to output 0 or 1 levels, the following must be true:  
A. The information must have been stored within the buffer  
B. The output enable must be true  
C. The output enable must be false  
D. The signal  $\overline{OE}$  must be at a logic 1
34. The dual of the statement  $(A+1)=1$  is  
A.  $A.1 = A$   
B.  $A.0 = 0$   
C.  $A+A = A$   
D.  $A.A = 1$
35. A unique advantages feature of CMOS logic family is its  
A. use of NMOS circuits  
B. power dissipation in nanowatt range  
C. speed  
D. dependent on frequency for power dissipation
36. If a multiplexer circuit has four input lines and a single output line, how many bits of line select code are required?



- A. 0                      C. 1  
B. 2                      D. 4
37. The pulse-width of an active high clock pulse is the time the:
- A. Data will be held on the output after a clock signal  
B. Data must be held on the input after a clock edge  
C. Clock must be high  
D. Clock must be low
38. Which of the following circuits is used for production of delays?
- A. Astable multivibrator  
B. Bistable multivibrator  
C. Monostable multivibrator  
D. Schmitt-trigger
39. What is the decimal equivalent of the hexadecimal number 3E8?
- A. 576                      C. 1000  
B. 1025                      D. 8350
40. The typical number of bits in a static memory location is:
- A. 1                      C. 2  
B. 8                      D. 16
41. Like a latch, the flip-flop belongs to a category of logic circuits known as
- A. monostable multivibrators  
B. astable multivibrators  
C. bistable multivibrators  
D. None of these
42. According to the absorptive Laws of Boolean algebra, expression  $(A+AB)$  equals
- A. A                      C. B  
B. AB                      D. A
43. The invalid state of a NAND latch occurs when
- A. S=1, C=0  
B. S=0, C=1  
C. S=1, C=1  
D. S=0, C=0
44. The purpose of the processing function in a system is:
- A. To store information that is needed later  
B. To generate outputs on the basis of present and past inputs  
C. To get information into, act on it, and output it from the system  
D. None of these
45. The main advantage of CMOS logic family over the TTL family is its
- A. much reduced power  
B. increased speed of operation  
C. extremely low cost  
D. series base resistor
46. A static memory generally contains:
- A. No decoders  
B. Row decoders  
C. Column decoders  
D. Both row and column decoders
47. The set-up time for a D-type flip-flop is the time that the:
- A. Data will be held on the output after a clock signal  
B. Data must be held on the input after a clock edge  
C. Clock must be high  
D. Data must be stable prior to a clock edge
48. A flip-flop which can have an uncertain output state is:
- A. J-K flip-flop  
B. S-C flip-flop  
C. D flip-flop  
D. T flip-flop
49. The chief advantage of Schottky TTL logic family is its least
- A. power dissipation  
B. propagation delay  
C. fan-in  
D. noise immunity
50. An encoder is an MSI circuit that:
- A. Provides for delivering one of two or more inputs to an output  
B. Selects a given output line based on a binary input code  
C. Provides an output code that corresponds to which of a set of input lines is true  
D. Provides a storage of a certain number of binary bits
- END